Abdullah Al Zahid

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SUMMARY

Transportation engineer with a master's degree and four years across municipal transportation planning at the City of Toronto, GO rail service planning at Metrolinx and a consulting placement at Mott MacDonald. Reviews traffic and transportation studies, runs traffic and rail simulation, and builds the Python and GIS tooling behind the analysis.

EXPERIENCE

Assistant Planner, Transportation Planning

City of Toronto | January 2024 – present

- Review development applications and the transportation studies filed with them against City standards and Official Plan policy, and recommend conditions where something is missing.
- Analyze travel and network data, including several years of the Transportation Tomorrow Survey, to establish where capacity falls short and what that means for future service and investment.
- Build and maintain the road and right-of-way datasets the division depends on, using GIS and Python, including a ROW-width dataset covering all city roads, built with QGIS, MapBox and JavaScript.
- Built the Transportation Data Dashboard, replacing a report rebuilt by hand each quarter with one validated source the division reports from, and saw the change through to adoption.

Transportation Planner, Simulation and Modelling

Metrolinx, Service Design | February 2023 – January 2024

- Ran OpenTrack simulations of GO rail services to establish what proposed operating and infrastructure changes would do to run times, reliability and recovery before they were taken forward.
- Assessed the service effect of temporary and permanent slow orders, the kind of restriction that construction on an operating railway imposes.
- Built the Crew Variance Tool, a Python application that read large JSON operations files and produced summaries of speed profiles and run-time variance across subdivisions.
- Set up a command line workflow so temporary and permanent slow orders could be pre-simulated in batches, cutting turnaround on scenario requests from days to hours.

Graduate Transportation Planner

Mott MacDonald | January – February 2023

- Short consulting placement on Ontario Line work. Ran traffic simulations to estimate how construction staging would affect surrounding streets, and prepared the supporting material for the project team.

Graduate Research Assistant

University of Alberta | January 2020 – August 2022

- Master's research on how road networks perform during emergency evacuations, using macroscopic (LWR) and agent-based (MATSim) models.
- Worked with large travel datasets and wrote up results for readers without a modelling background.

SKILLS

Synchro and VISSIM; OpenTrack and MATSim. Python (Pandas, NumPy), SQL and advanced Excel. ArcGIS Pro, QGIS and MapBox. AutoCAD and Civil 3D. Travel data analysis, including the Transportation Tomorrow Survey. Technical reports and presentations.

EDUCATION

MSc, Transportation Engineering — University of Alberta, 2022

BSc, Civil Engineering — Bangladesh University of Engineering and Technology, 2018

BSc, Computer Science — University of British Columbia, in progress (2025–2027)